# Sensitivity Analysis and Robust Estimation in Additive IV

Ben Deaner and Yuanqi Zhang *

October 26, 2026

**Abstract**

Recent work shows that in additive instrumental variables (IV) analysis, the robustness to some small failure of the identifying conditional moment restriction depends on the strength of the a priori smoothness or functional-form assumptions on the structural relationship between outcomes and treatments. In this work we develop methods for empirical sensitivity analysis that can help applied researchers assess the terms of this trade-off. We suggest a related estimator targeting the midpoint of an identified interval, which minimizes worst-case scalar error under a relaxation of the point-identifying assumptions. In effect, this estimator applies a novel regularization scheme, one that is motivated by the possibility that the IV moment condition is misspecified. We apply our methods to the consumer demand data previously analyzed in Blundell et al. (2007) and Horowitz (2011).

Instrumental variables (IV) analysis provides a means of empirically evaluating causal effects when treatment is endogenous. However, the validity of this approach rests on strong assumptions: the instruments must be assigned as if by a randomized experiment and they must not directly impact outcomes. Additive IV models (e.g., linear IV or the nonparametric IV model of Newey and Powell (2003) and Ai and Chen (2003)) also require that the outcome model has a particular additive structure, which may be difficult to motivate using setting-specific knowledge or economic theory. As a

*Deaner: University College London and CeMMAP, bendeaner@gmail.com; Zhang: University College London and CeMMAP uctpyqz@ucl.ac.uk.

result, applied researchers typically argue not that their IV model is exactly correctly specified, but rather that it holds to a close approximation.[1]

Even some small degree of misspecification may be consequential. Prior work shows that in additive IV models in particular, statistical inference can be highly misleading under misspecification, but this depends crucially on some empirically estimable quantities and on certain modeling choices made by the researcher. In particular, Deaner (2025) demonstrates that the sensitivity of IV inference to misspecification depends crucially on the object of interest (e.g., a weighted average partial effect) and the flexibility of the additive IV model considered, with more flexible models associated with greater sensitivity. Additionally, the classic work of Bound et al. (1995) shows that the bias in linear IV estimates resulting from misspecification is amplified if the instruments are weakly correlated with treatments.

Motivated by these concerns, the present work provides methods for empirical sensitivity analysis in flexible additive IV models, particularly models in which the causal relationship between treatments and outcomes (the 'structural function') is not assumed to take any particular functional form. Our approach also yields a related point estimator, which is 'robust' in the sense that it targets an estimand with minimal worst-case scalar error under misspecification. We show that the robustness of the estimator acts as a form of 'regularization' (see e.g., Chen and Pouzo (2012)) and derive conditions for consistency under correct specification as the misspecification tolerance shrinks.

We quantify sensitivity to misspecification similarly to Deaner (2025), which in turn follows the conceptual approach of Masten and Poirier (2018). Methods for sensitivity analysis are increasingly popular in applied economics and include local methods (e.g., Andrews et al. (2017), Armstrong and Kolesar (2018), Bonhomme and Weidner (2022)), and global methods (e.g., Altonji et al. (2005), Masten and Poirier (2018), Oster (2019)). The present work takes a global approach. To be more precise, we assess sensitivity by estimating the identified set when the point-identifying assumptions are relaxed. Additive IV models are point-identified by a conditional moment equality. If we relax this conditional moment equality so that it holds only up to some approximation error $b > 0$, then objects of interest are (in general) only partially identified. We say that an empirical finding is robust to misspecification if it holds up when one allows for some small degree of misspecification, that is, if it

[1] See Conley et al. (2008) for further discussion.

is implied by the identified set for some small but non-zero $b$. For example, if the identified set for an average partial effect (APE) contains only positive values, then the finding that the APE is positive is robust to misspecification.

In practice, the identified set is not known a priori, and must be estimated empirically: we provide methods to do so. The mid-point of the identified set for an object of interest is, by construction, an estimand with the lowest worst-case bias under the assumptions that define the set. Our robust estimator is an estimate of this mid-point. Both the set estimation and robust estimator can be evaluated by linear programming at little computational cost. This approach also allows researchers to easily impose additional linear restrictions, like smoothness or shape restrictions, without complicating estimation. We also illustrate subsampling-based uncertainty summaries (Politis and Romano, 1994); their coverage requires additional distributional and bias conditions discussed in Section 4.

We show that the set estimation problem is locally stable in nonparametric models when the discrepancy bound is separated from the feasibility boundary. Practically speaking, this means that the identified set can be estimated at a rate that does not depend on any 'sieve measure of ill-posedness' or 'sieve modulus of continuity' (Chen and Pouzo (2012)). This is because the relaxation of the conditional moment restriction effectively acts to regularize the estimation problem. This property is inherited by the robust estimator, with a stability factor that can deteriorate as the discrepancy bound approaches the feasibility boundary. The discrepancy $b$ specified by the researcher acts to control the degree of regularization. Unlike standard methods that may use the sieve dimension, Tikhonov regularization parameter, or quasi-Bayes prior (see Kankanala (2025)), $b$ is interpretable as the maximum amount of misspecification the researcher is unwilling to rule out, and we provide guidance for calibrating this quantity.

We apply our methods to the empirical setting shared by Blundell et al. (2007) and Horowitz (2011) who use nonparametric IV methods to estimate shape-invariant Engel curves using data from The British Family Expenditure Survey. We argue that in this setting instruments are unlikely to be fully exogenous, in which case the NPIV moment restriction will not hold exactly. We use our methodology to assess which features of the structural Engel curve for food can be inferred robustly.

# 1 Modeling Misspecification in Additive IV

Let $Y$ be an outcome, $X$ a treatment or vector of treatments, and $Z$ a vector of instrumental variables (IVs). Let $S_X$ and $S_Z$ denote their respective treatment and instrument supports. We consider models of the following form

$$Y = h_0(X) + \varepsilon. \tag{1.1}$$

The function $h_0$ is understood to capture a causal relationship between the outcome $Y$ and treatment/s $X$. As a leading example, let $Y(x)$ be the 'potential outcome' under a fixed counterfactual treatment level $x$ and define $h_0(x) := E[Y(x)]$, then $h_0(x)$ is the average potential outcome from treatment level $x$. Interest is in either $h_0$ itself or perhaps more commonly, some linear functional of $h_0$, for example the average causal effect of a 10% increase in the treatment level, which is, in the case of $h_0(x)$ the average potential outcome, $E[h_0(1.1X) - h_0(X) \mid X \in S_*]$, where $S_* = \{x \in S_X : 1.1x \in S_X\}$ and $P(X \in S_*) > 0$. This restricts the comparison to treatment values with common support.

The corresponding additive IV model is obtained by imposing the restriction that the additive 'structural residual' $\varepsilon$ is conditionally mean zero given the instrument/s $Z$, that is,

$$E[Y - h_0(X) \mid Z] = 0, \tag{1.2}$$

where the equality should be understood to hold with probability 1 (we follow this convention throughout the paper). Under a condition known as 'statistical completeness', the conditional moment restriction above identifies $h_0$, even in the absence of any further restrictions on this object (Newey and Powell (2003), Ai and Chen (2003)). If the conditional moment restriction (1.2) holds then we say the model is 'correctly specified'. We do not consider the model correctly specified if the restriction holds for *some* $h_0$, only if it holds for the particular $h_0$ of interest (e.g., the average potential outcome function). The desire to estimate a particular causally-interpretable object is what motivates the use of IV methods over say, regression of $Y$ on $X$.

Deaner (2025) provides a causal interpretation of the model above (see Appendix A.1 in that paper for details). He shows that when $h_0(x)$ is an average potential outcome from counterfactual treatment level $x$, the conditional moment restriction above follows from instrumental validity restrictions similar to those of Angrist et al.

(1996), namely an exclusion restriction and random assignment of the instrument. However, unlike in Angrist et al. (1996), the additive structure in the model above necessitates a strong additional assumption 'exogenous effects', which states that any heterogeneity in treatment effects is jointly mean independent of both the instruments and treatments. Deaner (2025) shows that if any one of these conditions fails, then the model is generally misspecified in the sense that (1.2) does not hold.

In order to examine sensitivity to misspecification Deaner (2025) suggests one relax the conditional moment equality. Define the function $u_0$ by

$$u_0(Z) := E[Y - h_0(X) \mid Z].$$

Using this notation, the conditional moment equality (1.2) states that $u_0(Z) = 0$. We can relax this in a number of ways. As a leading example, Deaner (2025) considers the conditional moment *inequality*:

$$|u_0(Z)| \leq b, \tag{1.3}$$

where the inequality should be understood to hold with probability 1.[2] The positive scalar $b$ represents a bound on the degree of misspecification, with $b = 0$ corresponding to correct specification. A small but strictly positive $b$ means the specification is approximately correct: the conditional moment equality (1.2) may not hold, but it must *almost hold* in that $E[Y - h_0(X) \mid Z]$ must be uniformly close to zero.

The inequality (1.3) provides a means of modeling misspecification that is agnostic about the precise source of that misspecification, e.g., violation of exogenous effects or of the exclusion restriction. Deaner (2025) characterizes $u_0$ under various sources of misspecification (e.g., violation of the exclusion restriction) and provides corresponding interpretations of the bound $b$. As a leading example, suppose that the IV exclusion restriction fails so that there is an additive direct effect of the instrument $Z$ on the outcome $Y$. Let $\gamma_0$ capture this direct effect so that the potential outcome from counterfactual treatment level $x$ and IV level $z$ is

$$Y(x, z) = h_0(x) + \gamma_0(z) + \varepsilon.$$

[2]The analysis in Deaner (2025) is somewhat more general. He imposes a bound $b$ on the norm of $u_0$ and allows for general choices of this norm. The restriction $|u_0(Z)| \leq b$ corresponds to a bound on the essential supremum norm and is thus a special case. Deaner (2025) also considers additional a priori restrictions on $u_0$, for example, that $E[u_0(Z)] = 0$.

Normalize $E[\gamma_0(Z)] = 0$ and suppose the instrument is randomly assigned so that $E[\varepsilon|Z] = 0$. Then we have $u_0 = \gamma_0$. The average direct effect of setting the instrument to level $z$ is then $E[Y(x,z) - Y(x,Z)] = \gamma_0(z)$, and so (1.3) imposes that the average direct effect has magnitude smaller than $b$.

## 1.1 An Approach to Sensitivity Analysis

The relaxed condition (1.3) is central to our approach to sensitivity analysis. An inference made under the assumption (1.2) is 'robust to misspecification' if it remains valid under the relaxation (1.3) for some small $b > 0$. That is, one does not need to believe the conditional moment restriction holds exactly for the finding to be valid and instead it suffices that the condition only holds approximately. If it remains valid under (1.3) for a larger $b$ then it is relatively more robust.

To formalize this, consider that, under statistical completeness, the moment equality (1.2) point identifies $h_0$, whereas (1.3) for a fixed $b > 0$ generally only partially identifies $h_0$. The corresponding identified set captures everything that could be learned about $h_0$ from the data under the assumption (1.3). If say, every element of the identified set is monotone, then the finding that $h_0$ is monotone is robust. Under this approach, empirical sensitivity analysis amounts to the problem of estimating (and performing inference on) the identified set. This may be the identified set for $h_0$ or for a particular functional of interest. Note that this approach to sensitivity analysis is estimator-free: rather than assess say, the bias of a particular IV estimator under misspecification, we directly assess the robustness of conclusions one might draw from the data.

We now formally define the identified set. Let $\mathcal{H}$ be a set of functions assumed to contain $h_0$. This set may incorporate additional a priori restrictions on $h_0$, for example, if the researcher believes $h_0$ is convex then one can impose this assumption by letting $\mathcal{H}$ contain only convex functions. The identified set, given the assumptions that $h_0 \in \mathcal{H}$ and (1.3) holds, is then formally defined by

$$\Theta_b := \left\{ h \in \mathcal{H} : \left| E[Y - h(X) \mid Z] \right| \leq b \right\}. \tag{1.4}$$

The identified set varies with the choice of $b$ which bounds the degree of misspecification. A larger value of $b$ entails a larger identified set and thus under a larger $b$ less

can be learned from the data about $h_0$. By varying $b$ and inspecting the identified set (or in practice, an estimate of the set), one can assess which features of $h_0$ can be inferred under relaxations of the conditional moment restriction.

In practice, researchers may not be interested in the whole structural function $h_0$ but a functional thereof. Recall the example in the previous subsection in which interest is in the average causal effect of a 10% increase in the treatment level $E[h_0(1.1X) - h_0(X) \mid X \in S_*]$. This object is a scalar functional of $h_0$, it is a linear functional. Our empirical methods are designed specifically for linear functionals. Another important linear functional is the value of $h_0$ at a fixed point $x$, which is simply $h_0(x)$. Let $\mathbb{L}$ be a scalar functional defined on $\mathcal{H}$. That is, $\mathbb{L}[h]$ takes the function $h \in \mathcal{H}$ and returns a scalar. The identified set for $\mathbb{L}[h_0]$ is the image of $\Theta_b$ under $\mathbb{L}$, denoted $\mathbb{L}[\Theta_b]$, and it can be defined directly by

$$\mathbb{L}[\Theta_b] := \Big\{ \mathbb{L}[h] : \ h \in \mathcal{H} \ \text{ and } \ \big|E[Y - h(X) \mid Z]\big| \leq b \Big\}. \tag{1.5}$$

A key finding in Deaner (2025) is that the additional assumptions a researcher places on $h_0$ (embodied in the restrictiveness of $\mathcal{H}$) can be highly consequential for the sensitivity to misspecification. Stronger restrictions on $h_0$ correspond to lower sensitivity. More precisely, the rate at which the diameter of $\Theta_b$ shrinks as $b$ approaches zero depends crucially on $\mathcal{H}$, and if only very weak restrictions are placed on $\mathcal{H}$ then the diameter of the identified set does not shrink to zero with $b$. This phenomenon may or may not also apply to $\mathbb{L}[\Theta_b]$, depending on the particular linear functional of interest $\mathbb{L}$ and the joint distribution of $X$ and $Z$. In sum, there is a trade-off between the flexibility of the a priori restrictions placed on $h_0$ and the sensitivity to misspecification of the conditional moment equality. Estimation of $\Theta_b$ and $\mathbb{L}[\Theta_b]$ for different choices of $\mathcal{H}$ may aid researchers in making this trade-off.

## 1.2 A tractable characterization of the identified set

Let $\mathcal{B}_X = C_b(S_X)$ denote the bounded continuous real-valued functions on the support $S_X$, equipped with the supremum norm $\|h\|_\infty = \sup_{x \in S_X} |h(x)|$. Let $\mathcal{D} \subseteq \mathcal{B}_X$ be a linear domain on which the chosen shape operators are defined. For example, derivative restrictions require a suitably differentiable domain. Let $S_H \subseteq \mathbb{R}^p$ index the shape restrictions, and let $\mathbb{T}$ be a known linear operator from $\mathcal{D}$ to $\mathbb{R}^d$-valued

functions on $S_H$. For a known vector function $C : S_H \to \mathbb{R}^d$, take

$$\mathcal{H} = \{h \in \mathcal{D} : \mathbb{T}[h](x) \leq C(x),\ x \in S_H\}. \tag{1.6}$$

Vector inequalities hold componentwise. Often $S_H = S_X$, after a normalization of the treatment support. For example, the restrictions $0 \leq h \leq 1$ and $|h''| \leq c$ correspond to $\mathbb{T}[h] = (h, -h, h'', -h'')'$ and $C = (1, 0, c, c)'$ on a compact interval, with $\mathcal{D} = C^2(S_X)$.

The set $\mathcal{H}$, and therefore $\Theta_b$, is convex. For a scalar linear functional $\mathbb{L}$, the image $\mathbb{L}[\Theta_b]$ is an interval whenever it is nonempty. Its endpoints need not be attained unless additional compactness or closedness conditions hold. Write them as

$$\underline{\theta}_{\mathbb{L}}(b) = \inf_{h \in \mathcal{H}} \mathbb{L}[h] \quad \text{s.t. } |E[Y - h(X) \mid Z]| \leq b, \tag{1.7}$$

$$\bar{\theta}_{\mathbb{L}}(b) = \sup_{h \in \mathcal{H}} \mathbb{L}[h] \quad \text{s.t. } |E[Y - h(X) \mid Z]| \leq b. \tag{1.8}$$

These are linear programs because the absolute-value inequality is equivalent to two linear inequalities. Whenever we discuss their midpoints, we require a nonempty set and finite endpoints.

For point evaluation $\mathbb{L}[h] = h(x)$, write

$$\underline{\theta}(x; b) = \inf_{h \in \Theta_b} h(x), \qquad \bar{\theta}(x; b) = \sup_{h \in \Theta_b} h(x). \tag{1.9}$$

These are the lower and upper envelopes of $\Theta_b$. They are useful for displaying pointwise identified intervals, but they do not describe all joint restrictions on the structural function. For example, an arbitrary function lying between the envelopes need not belong to $\Theta_b$. We suppress the argument $b$ when it is fixed.

## 1.3 Robust point estimands

Our analysis motivates a robust point estimator of the structural function $h_0$ or a scalar linear functional $\mathbb{L}[h_0]$. Our approach is to estimate the midpoint of the corresponding identified interval. Define

$$\theta^{\text{mid}}_{\mathbb{L}}(b) = \tfrac{1}{2}\{\underline{\theta}_{\mathbb{L}}(b) + \bar{\theta}_{\mathbb{L}}(b)\}, \qquad \theta^{\text{mid}}(x; b) = \tfrac{1}{2}\{\underline{\theta}(x; b) + \bar{\theta}(x; b)\}. \tag{1.10}$$

An estimate of $\theta_{\mathbb{L}}^{\text{mid}}$ is our robust estimate of $\mathbb{L}[h_0]$, and an estimate of $\theta^{\text{mid}}(x)$ is our robust estimate of $h_0(x)$. These targets are robust in the sense that they minimize worst-case squared scalar loss under the assumptions defining the identified set. This decision-theoretic motivation for estimating the midpoint of an identified interval is also studied by Song (2014) and Dominitz and F. Manski (2017). We first explain the oracle decision problem, then relate it to the minimax risk of feasible estimators. Throughout this discussion, the misspecification tolerance $b$ is fixed.

Consider a deterministic estimate $c$ of $\mathbb{L}[h_0]$, with squared loss $(c - \mathbb{L}[h_0])^2$. If all that is known about $h_0$ is that it lies in $\Theta_b$, the worst possible loss is $\sup_{h\in\Theta_b}(c-\mathbb{L}[h])^2$. The unique minimizer is the midpoint $\theta_{\mathbb{L}}^{\text{mid}}$. Let $d_{\mathbb{L}} = \bar{\theta}_{\mathbb{L}} - \underline{\theta}_{\mathbb{L}}$, then the midpoint achieves the minimized loss

$$\inf_{c\in\mathbb{R}} \sup_{h\in\Theta_b} (c - \mathbb{L}[h])^2 = d_{\mathbb{L}}^2/4. \tag{1.11}$$

This is an oracle decision because the identified interval depends on the unknown distribution of observables.

For a formal statistical argument, let a model $m$ consist of a joint distribution of $(X, Z, \varepsilon)$ and a structural function $h_m$. Together with $Y = h_m(X)+\varepsilon$, these determine a distribution $P_m$ of observables $(X, Z, Y)$. Let $\mathcal{M}$ be the class of models satisfying the restrictions that define the identified set: $h_m \in \mathcal{H}$ and $|E_{P_m}[Y - h_m(X) \mid Z]| \leq b$ almost surely. Let $\mathcal{P}$ be a class of distributions of observables, and suppose the true model belongs to $\mathcal{M}$ and its observable distribution $P_0$ belongs to $\mathcal{P}$. Write $\Theta_b(P)$, $\theta_{\mathbb{L}}^{\text{mid}}(P)$, and $d_{\mathbb{L}}(P)$ to make the dependence on an observable distribution $P$ explicit. For this argument, $\mathcal{H}$ and $\mathbb{L}$ are fixed.

Suppose $\hat{c}_n$ is a measurable estimator based on $n$ independent observations from $P_m$. Its maximum expected squared loss over the models under consideration is

$$R_n(\hat{c}_n; \mathcal{P}) := \sup_{m\in\mathcal{M}:\, P_m\in\mathcal{P}} E_{P_m^{\otimes n}}[(\hat{c}_n - \mathbb{L}[h_m])^2]. \tag{1.12}$$

The subscript $P_m^{\otimes n}$ on the expectation indicates that this is the mean when the observations are distributed iid according the probability distribution $P_m$. Every estimator

with finite maximum risk satisfies

$$\begin{aligned} R_n(\hat{c}_n;\mathcal{P}) &\geq \sup_{m\in\mathcal{M}:\, P_m=P_0} \left(E_{P_0^{\otimes n}}[\hat{c}_n] - \mathbb{L}[h_m]\right)^2 \\ &= \sup_{h\in\Theta_b(P_0)} \left(E_{P_0^{\otimes n}}[\hat{c}_n] - \mathbb{L}[h]\right)^2 \\ &\geq \inf_{c\in\mathbb{R}} \sup_{h\in\Theta_b(P_0)} (c - \mathbb{L}[h])^2 = \frac{d_{\mathbb{L}}(P_0)^2}{4}. \end{aligned} \tag{1.13}$$

The first inequality restricts attention to models with the same observable distribution $P_0$ and applies Jensen's inequality. These models are observationally equivalent, so the distribution of $\hat{c}_n$ is the same under all of them. The first equality holds because their structural functions are precisely the elements of $\Theta_b(P_0)$: for any $h$ in that set, defining $\varepsilon = Y - h(X)$ under $P_0$ constructs a compatible model. The final equality is (1.11). The risk lower bound also holds trivially for estimators with infinite maximum risk.

The minimum in (1.13) is achieved by setting $c = \theta_{\mathbb{L}}^{\text{mid}}$ which yields minimum $\frac{1}{2}(\bar{\theta}_{\mathbb{L}} - \underline{\theta}_{\mathbb{L}})$. That is, $\theta_{\mathbb{L}}^{\text{mid}}$ is an oracle estimate of $\mathbb{L}[h_0]$ that achieves minimax risk under the assumptions that define the identified set. If $\hat{\theta}_{\mathbb{L}}^{\text{mid}}$ is a feasible estimator that converges in mean square to $\theta_{\mathbb{L}}^{\text{mid}}$ uniformly over $P \in \mathcal{P}$, then its maximum risk converges to the minimax risk. Similar reasoning applies for a uniformly mean-square consistent estimator of $\theta^{\text{mid}}(x)$.[3]

# 2 Estimation

We approximate the population linear programs using a sieve, estimated conditional means, and finite constraint grids. Let $\Phi_n$ be a length-$K_n$ column vector of basis functions in $\mathcal{D}$, and approximate $h$ by $\Phi_n'\beta$. Write $g_0(z) = E[Y \mid Z = z]$ and $\Pi_n(z) = E[\Phi_n(X) \mid Z = z]$. Nonparametric regressions provide estimates $\hat{g}_n$ and $\hat{\Pi}_n$, yielding the estimated residual $\hat{g}_n(z) - \hat{\Pi}_n(z)'\beta$.

Choose finite, nonempty grids $\widetilde{S}_Z \subseteq S_Z$ and $\widetilde{S}_H \subseteq S_H$. Define the feasible con-

[3]The restriction of uniformity over $\mathcal{P}$ is stronger than needed, uniformity over the subset of distributions in $\mathcal{P}$ which are compatible with models in $\mathcal{M}$ would suffice.

straints

$$|\hat{g}_n(z) - \hat{\Pi}_n(z)'\beta| \le b, \qquad z \in \widetilde{S}_Z, \tag{2.1}$$

$$\mathbb{T}[\Phi_n'](x)\beta \le C(x), \qquad x \in \widetilde{S}_H. \tag{2.2}$$

Here $\mathbb{T}[\Phi_n']$ is the $d \times K_n$ matrix obtained by applying $\mathbb{T}$ to each basis function. A grid generally only approximates the continuum restrictions. Checking spline knots is sufficient for a constraint only when the relevant piecewise polynomial attains its extrema there; this is not true for arbitrary spline value constraints.

Let $\mathbb{L}[\Phi_n]$ denote the column vector of functional values of the basis elements. The endpoint estimators are

$$\underline{\hat{\theta}}_{\mathbb{L}}(b) = \inf_{\beta \in \mathbb{R}^{K_n}} \mathbb{L}[\Phi_n]'\beta \quad \text{s.t. (2.1)–(2.2)}, \tag{2.3}$$

$$\hat{\bar{\theta}}_{\mathbb{L}}(b) = \sup_{\beta \in \mathbb{R}^{K_n}} \mathbb{L}[\Phi_n]'\beta \quad \text{s.t. (2.1)–(2.2)}. \tag{2.4}$$

For point evaluation the objective is $\Phi_n(x)'\beta$; denote the resulting endpoints by $\underline{\hat{\theta}}(x;b)$ and $\hat{\bar{\theta}}(x;b)$. The theory treats $\mathbb{L}$ as given; if its evaluation requires estimating an unknown distribution, the additional objective-estimation error must also be controlled. Each program has $K_n$ variables and $2|\widetilde{S}_Z| + d|\widetilde{S}_H|$ scalar inequality constraints. The midpoint estimators are

$$\hat{\theta}^{\text{mid}}_{\mathbb{L}}(b) = \tfrac{1}{2}\{\underline{\hat{\theta}}_{\mathbb{L}}(b) + \hat{\bar{\theta}}_{\mathbb{L}}(b)\}, \qquad \hat{\theta}^{\text{mid}}(x;b) = \tfrac{1}{2}\{\underline{\hat{\theta}}(x;b) + \hat{\bar{\theta}}(x;b)\}. \tag{2.5}$$

An empty sample program should be reported as infeasible at the chosen $b$; its midpoint is undefined. Theorem 3.2 gives conditions ensuring feasibility with probability approaching one.

## 2.1 Calibration of $b$ and choosing tuning parameters

The bound $b$ in (1.3) determines the size of the identified set. We suggest the researcher estimate the identified set for a range of choices for $b$ corresponding to mild, moderate, and severe misspecification. To determine whether $b$ corresponds to say, 'mild' misspecification, it is helpful to compare $b$ to an estimable reference quantity. This approach is common in the empirical sensitivity literature. For example, Masten

and Poirier (2018) compare the degree of selection on unobservables to selection on observables, which is estimable. See also Imbens (2003), Altonji et al. (2005), and Oster (2019).

We suggest calibrating $b$ against the residual variation in the outcome. Consider the following decomposition:

$$Y = E[h_0(X)|Z] + u_0(Z) + \eta$$

Under the additive direct-effect interpretation in Section 1, the first two terms capture the indirect and direct components associated with the instrument. More generally, $u_0$ aggregates departures from the conditional moment restriction. The residual $\eta$ absorbs the effect of all other factors that influence $Y$. The variation in $\eta$ is a useful reference quantity against which to calibrate the bound $b$ on the magnitude of $u_0(Z)$. We measure the variation using a difference in quantiles.

Let $q_\eta(\alpha)$ denote the $\alpha$ quantile of $\eta$. We say $b$ is small if it is equal to $q_\eta(0.5 + \tau) - q_\eta(0.5 - \tau)$ for a small $\tau$. For example, $b = q_\eta(0.6) - q_\eta(0.4)$ corresponds to $\tau = 0.1$. In applications, the researcher can report identified sets for several values of $\tau$, corresponding to mild, moderate, and severe departures from exact instrument validity. Note that $\eta = Y - E[Y|Z]$ is a reduced-form residual and thus we can estimate $\eta$ and its quantiles in a first stage.

# 3 Theoretical Analysis

Write $g_0(z) = E[Y \mid Z = z]$ and $\mathbb{A}[h](z) = E[h(X) \mid Z = z]$. We use bounded continuous versions of these conditional means on $S_Z$. The operator $\mathbb{A} : \mathcal{B}_X \to \mathcal{B}_Z$, where $\mathcal{B}_Z = C_b(S_Z)$, is linear and satisfies $\|\mathbb{A}[h]\|_\infty \leq \|h\|_\infty$. Correct specification is the equation

$$g_0 = \mathbb{A}[h_0]. \tag{3.1}$$

All stochastic results concern an i.i.d. sample $\{(Y_i, X_i, Z_i)\}_{i=1}^n$. Function classes are taken to be separable for the suprema below; alternatively, probability statements can be read in outer probability. For a bounded linear functional on $\mathcal{B}_X$, write $\|\mathbb{L}\|_{\text{op}}$ for its supremum-norm operator norm. Point evaluation has norm one. Derivative evaluation is not generally bounded in this norm, and the results for bounded functionals do not automatically cover it.

## 3.1 Midpoint estimation as a regularization scheme

If $\mathbb{A}$ is injective, correct specification identifies $h_0 = \mathbb{A}^{-1}[g_0]$. In an ill-posed NPIV problem this inverse is discontinuous on its range. The midpoint instead targets the center of the identified interval at a positive discrepancy bound. The following result describes its local stability, including perturbations of both $g_0$ and $\mathbb{A}$.

Let $\widetilde{g}$ and $\widetilde{\mathbb{A}}$ be a bounded function and a linear operator, respectively. A tilde on an endpoint denotes its value with these objects replacing $g_0$ and $\mathbb{A}$, at the same bound $b$ and with the same class $\mathcal{H}$. Define

$$\eta = \sup_{h \in \mathcal{H}} \|(\widetilde{g} - \widetilde{\mathbb{A}}[h]) - (g_0 - \mathbb{A}[h])\|_\infty.$$

**Proposition 3.1** (Stability of endpoints and midpoints)**.** *Suppose $\mathcal{H}$ is convex, $\eta < \infty$, and $h^* \in \mathcal{H}$ satisfies $\|g_0 - \mathbb{A}[h^*]\|_\infty \leq \varepsilon_0$. For $b > \varepsilon_0 + \eta$, suppose the population endpoints for $\mathbb{L}$ are finite. Then the perturbed endpoints are also finite and*

$$|\widetilde{\bar{\theta}}_{\mathbb{L}}(b) - \bar{\theta}_{\mathbb{L}}(b)| \leq \frac{\eta}{b - \varepsilon_0}\{\bar{\theta}_{\mathbb{L}}(b) - \mathbb{L}[h^*]\},$$
$$|\widetilde{\underline{\theta}}_{\mathbb{L}}(b) - \underline{\theta}_{\mathbb{L}}(b)| \leq \frac{\eta}{b - \varepsilon_0}\{\mathbb{L}[h^*] - \underline{\theta}_{\mathbb{L}}(b)\}.$$

*Consequently,*

$$|\widetilde{\theta}^{\text{mid}}_{\mathbb{L}}(b) - \theta^{\text{mid}}_{\mathbb{L}}(b)| \leq \frac{\eta}{2(b - \varepsilon_0)}\{\bar{\theta}_{\mathbb{L}}(b) - \underline{\theta}_{\mathbb{L}}(b)\}.$$

*The same statements hold for point evaluations, uniformly in $x$ if the corresponding widths are uniformly bounded.*

Under correct specification one can take $h^* = h_0$ and $\varepsilon_0 = 0$. The stability bound then contains the ratio of identified-set width to $b$. This is a local Lipschitz bound, not a claim of differentiability. Its coefficient need not diverge as $b \downarrow 0$, since the width may shrink too. The rate at which the width shrinks depends on the restrictions embodied in $\mathcal{H}$ and on the conditional-expectation operator; see Deaner (2025).

## 3.2 Consistency and convergence rates

We make the dependence on $b$ explicit by writing $\underline{\theta}_{\mathbb{L}}(b)$, $\bar{\theta}_{\mathbb{L}}(b)$ and their sample counterparts. The same first-stage estimates, sieve, and grids are used for every $b$. Define

$$\mathcal{H}_n = \mathcal{H} \cap \operatorname{span}(\Phi_n), \qquad \Pi_n(z) = E[\Phi_n(X) \mid Z = z],$$

and the mesh widths

$$D_{H,n} = \sup_{x \in S_H} \min_{x' \in \widetilde{S}_H} \|x - x'\|_2, \qquad D_{Z,n} = \sup_{z \in S_Z} \min_{z' \in \widetilde{S}_Z} \|z - z'\|_2.$$

The basis functions are linearly independent. Let

$$\iota_n = 1 + \sup_{\beta \neq 0} \frac{\|\beta\|_2}{\|\Phi_n' \beta\|_\infty} < \infty.$$

**Assumption 3.1** (First-stage error)**.** For a deterministic sequence $a_n \downarrow 0$,

$$\eta_n := \|\hat{g}_n - g_0\|_\infty + \sup_{\beta:\, \Phi_n'\beta \in \mathcal{H}} \|(\hat{\Pi}_n - \Pi_n)'\beta\|_\infty = O_p(a_n).$$

**Assumption 3.2** (Feasible sieve approximation)**.** For a deterministic sequence $\kappa_n \downarrow 0$, each $h \in \mathcal{H}$ has an approximation $q_n[h] \in \mathcal{H}_n$ with $\|q_n[h] - h\|_\infty \leq \kappa_n$. In addition, there exist $h^s \in \mathcal{H}$, $\sigma > 0$, and $q_n^s \in \mathcal{H}_n$ such that, for all sufficiently large $n$,

$$\|q_n^s - h^s\|_\infty \leq \kappa_n, \qquad \mathbb{T}[q_n^s](x) \leq C(x) - \sigma \mathbf{1}, \quad x \in S_H.$$

The first part requires preservation of membership in $\mathcal{H}$, not the stronger inequalities $\mathbb{T}[q_n[h]] \leq \mathbb{T}[h]$. For paired upper and lower value restrictions, those stronger inequalities would force $q_n[h] = h$. The second part supplies a strictly shape-feasible sieve anchor. It does not require $h_0$ to be strictly inside the shape restrictions. For range and curvature restrictions, the constant function $h^s = 1/2$ is such an anchor when the curvature bound is positive and constants belong to the sieve.

**Assumption 3.3** (Grid regularity)**.** The vector function $C$ and each row of $\mathbb{T}[\Phi_n']$ are Lipschitz with constants at most $\xi_n$. With probability approaching one, $\hat{g}_n$ and the vector function $\hat{\Pi}_n$ are Lipschitz with constants at most $G_n$. Both mesh widths

tend to zero and

$$d_n := \iota_n \xi_n D_{H,n} + \iota_n G_n D_{Z,n} \longrightarrow 0.$$

Set $r_n = a_n + \kappa_n + d_n$. The next theorem uses boundedness of one population identified set, rather than requiring the shape inequalities at each individual point to bound the function there. In particular, no positivity restriction is imposed on the entries of $C$.

**Theorem 3.2** (A rate uniform in the discrepancy bound)**.** *Suppose Assumptions 3.1–3.3 hold. Fix $0 \leq \varepsilon_0 < \bar{b} < \infty$. Suppose there is $h^p \in \mathcal{H}$ with $\|g_0 - \mathbb{A}[h^p]\|_\infty \leq \varepsilon_0$, and*

$$M := \sup_{h \in \Theta_{\bar{b}}} \|h\|_\infty < \infty. \tag{3.2}$$

*Write $s(b) = \min\{1, b - \varepsilon_0\}$. For any deterministic sequence $\underline{b}_n \in (\varepsilon_0, \bar{b}]$ satisfying $r_n = o(s(\underline{b}_n))$, the sample programs are simultaneously nonempty and bounded for all $b \in [\underline{b}_n, \bar{b}]$ with probability approaching one. For every fixed $C_L < \infty$,*

$$\sup_{b \in [\underline{b}_n, \bar{b}]} s(b) \sup_{\|\mathbb{L}\|_{\mathrm{op}} \leq C_L} \max \left\{ |\hat{\underline{\theta}}_{\mathbb{L}}(b) - \underline{\theta}_{\mathbb{L}}(b)|, |\hat{\bar{\theta}}_{\mathbb{L}}(b) - \bar{\theta}_{\mathbb{L}}(b)| \right\} = O_p(r_n). \tag{3.3}$$

*The same bound holds for the midpoint errors. In particular, all three bounds hold uniformly over point evaluations $x \in S_X$.*

The implied stochastic bound in (3.3) is independent of $b$. On any fixed interval separated from $\varepsilon_0$, the unweighted rate is $O_p(r_n)$. Under correct specification, it is $O_p(r_n/b)$ as $b \downarrow 0$, provided $r_n = o(b)$. The theorem also applies to a data-dependent choice of $b$ lying in the stated interval with probability approaching one; it then compares the estimator with the population target evaluated at that same $b$. Convergence to a target at a fixed calibration value additionally requires control of calibration error.

The boundedness in (3.2) holds if $\mathcal{H}$ is bounded, but can also hold for an unbounded $\mathcal{H}$. It cannot be omitted without replacement: if the class contains an unbounded direction in the null space of $\mathbb{A}$, some endpoint objectives can be infinite. The proof derives boundedness of the grid-feasible sieve functions from (3.2) and the anchor, so no separate pointwise boundedness assumption is needed. It also allows zero or negative entries of $C$, including the lower range restriction $-h \leq 0$ in the application.

Dense grids make $d_n$ small. Increasing $K_n$ may reduce $\kappa_n$, but also changes $\iota_n$ and $\xi_n$, so the grid conditions must be checked jointly with the sieve dimension. The first-stage result below does not depend on $K_n$.

**Corollary 3.3** (Midpoint convergence as $b \downarrow 0$)**.** *Suppose the conditions of Theorem 3.2 hold with $g_0 = \mathbb{A}[h_0]$, $h_0 \in \mathcal{H}$, and $\varepsilon_0 = 0$. Suppose the supremum-norm closure $\overline{\mathcal{H}}$ is compact and $h_0$ is the unique element of this closure satisfying $\mathbb{A}[h] = g_0$. Define*

$$\omega(t) = \sup\{\|h - h_0\|_\infty : h \in \overline{\mathcal{H}},\ \|\mathbb{A}[h - h_0]\|_\infty \le t\}.$$

*Then $\omega(t) \to 0$ as $t \downarrow 0$, and*

$$\|\theta^{\mathrm{mid}}(\cdot; b) - h_0\|_\infty \le \omega(b).$$

*For any $b_n \downarrow 0$ with $r_n = o(b_n)$,*

$$\|\hat{\theta}^{\mathrm{mid}}(\cdot; b_n) - h_0\|_\infty \le \omega(b_n) + O_p(r_n/b_n) = o_p(1). \tag{3.4}$$

*Uniformly over $\|\mathbb{L}\|_{\mathrm{op}} \le C_L$, the analogous bound for $\hat{\theta}^{\mathrm{mid}}_{\mathbb{L}}(b_n)$ has bias term $C_L\omega(b_n)$. If $\omega(t) \le Ct^\alpha$ for some $\alpha > 0$, the choice $b_n \asymp r_n^{1/(1+\alpha)}$ gives the rate $O_p(r_n^{\alpha/(1+\alpha)})$.*

Completeness on the closure is sufficient for the uniqueness condition. Compactness and uniqueness are used to make the regularization bias vanish; neither follows merely from boundedness. Under misspecification, letting $b$ tend to zero need not target $h_0$ and may make the programs infeasible.

## 3.3 Primitive conditions for the first stage and sieve

For a length-$L_n$ column vector $\Psi_n$ on $S_Z$, let $Q_n = E[\Psi_n(Z)\Psi_n(Z)']$ and $\hat{Q}_n = n^{-1}\sum_{i=1}^n \Psi_n(Z_i)\Psi_n(Z_i)'$. Define

$$\hat{g}_n(z) = \Psi_n(z)'\hat{Q}_n^{-1} n^{-1} \sum_{i=1}^n \Psi_n(Z_i) Y_i, \tag{3.5}$$

$$\hat{\Pi}_n(z) = \left( n^{-1} \sum_{i=1}^n \Phi_n(X_i)\Psi_n(Z_i)' \right) \hat{Q}_n^{-1} \Psi_n(z). \tag{3.6}$$

Both $\Pi_n$ and $\hat{\Pi}_n$ are column vectors of length $K_n$. Set these estimates to zero if $\hat{Q}_n$ is singular; the assumptions below make that event asymptotically negligible.

For $s > 0$, let $m_s = \lceil s \rceil - 1$ and $\delta_s = s - m_s \in (0, 1]$. The Hölder norm $\|f\|_{C^s}$ bounds all partial derivatives through order $m_s$, including the function itself, and the $\delta_s$-Hölder seminorms of derivatives of order $m_s$. This convention includes Lipschitz functions at $s = 1$.

**Assumption 3.4** (Design and conditional smoothness)**.** The eigenvalues of $Q_n$ are uniformly bounded above and away from zero, and $S_Z$ is a bounded subset of $\mathbb{R}^{d_Z}$. There is a conditional density $f_{X|Z}$ with respect to a measure $\mu_X$ and an integrable envelope $\ell(x)$ such that

$$\|f_{X|Z}(x \mid \cdot)\|_{C^s} \leq \ell(x), \qquad \int \ell(x)\, d\mu_X(x) < \infty.$$

**Assumption 3.5** (Series approximation and entropy)**.** Define the population series projection by

$$P_n f(z) = \Psi_n(z)' Q_n^{-1} E[\Psi_n(Z) f(Z)].$$

There is $R_n(s) \downarrow 0$ such that $\|f - P_n f\|_\infty \leq \|f\|_{C^s} R_n(s)$. Also $\sup_z \|\Psi_n(z)\|_2 \leq \bar{\xi}_n$, $\Psi_n(z) \neq 0$, and $\alpha_n(z) = \Psi_n(z)/\|\Psi_n(z)\|_2$ is Lipschitz with constant $\ell_n \geq 1$. Finally,

$$\int_0^1 \sqrt{\log N(\mathcal{H}, \|\cdot\|_\infty, u)}\, du < \infty, \qquad \frac{\bar{\xi}_n^2 \log(2L_n)}{n} \longrightarrow 0.$$

Here $N$ denotes a covering number in the stated norm.

The entropy condition implies that $\mathcal{H}$ is totally bounded, and therefore uniformly bounded in supremum norm. Thus the first-stage argument does not require a separate condition that the shape restrictions imply pointwise bounds. The integrable envelope in Assumption 3.4 justifies differentiation under the integral and bounds the smoothness of $\mathbb{A}[h]$ by a constant times $\|h\|_\infty$.

**Assumption 3.6** (Reduced-form moments and growth)**.** The function $g_0$ has finite $C^s$ norm. For some $m > 2$,

$$\sup_{z \in S_Z} E[|Y - g_0(Z)|^m \mid Z = z] < \infty.$$

For some $v > 0$, as $L_n \to \infty$,

$$\bar{\xi}_n = O(\sqrt{L_n}), \quad \log \ell_n = O(\log L_n), \quad R_n(s) = O(L_n^{-v/d_Z}), \quad R_n(s) = O(L_n^{-1/2}),$$

and

$$\frac{\bar{\xi}_n^{2m/(m-2)} \log(2L_n)}{n} = O(1), \qquad \frac{L_n \log(2L_n)}{n^{1-2/m}} = O(1).$$

**Theorem 3.4.** *Under Assumptions 3.4–3.6, the estimates in (3.5)–(3.6) satisfy Assumption 3.1 with*

$$a_n = \sqrt{L_n \log(2L_n)/n} + L_n^{-v/d_Z}.$$

*The bound is uniform over all structural sieve dimensions $K_n$ and bases whose admissible functions belong to the fixed class $\mathcal{H}$.*

Balancing the two displayed terms gives $L_n \asymp (n/\log n)^{d_Z/(2v+d_Z)}$ and $a_n \asymp (\log n/n)^{v/(2v+d_Z)}$, provided this choice satisfies all the moment and growth conditions. The theorem concerns least squares first stages. For ridge first stages, ridge bias must also be included in $a_n$ unless it is negligible at this rate.

**Theorem 3.5** (Range- and curvature-preserving splines)**.** *Let $S_X = [x_-, x_+]$ be a nondegenerate compact interval, $c > 0$, and*

$$\mathcal{H}(c) = \{h \in C^2(S_X) : 0 \le h \le 1, \ \|h''\|_\infty \le c\}.$$

*Let $\Phi_n$ span the splines of a fixed order $q \ge 4$ with equally spaced interior knots and mesh width $\Delta_n \asymp K_n^{-1}$, where $K_n$ is the dimension of the spline space. Then Assumption 3.2 holds with $\kappa_n = O(K_n^{-1/2})$, using $h^s = q_n^s = 1/2$. The class has compact closure in the supremum norm and satisfies the entropy condition in Assumption 3.5.*

The order convention is degree plus one, so the theorem includes cubic splines. The curvature operators applied to this basis are continuous and piecewise polynomial, hence Lipschitz on the compact interval, as required for the grid comparison.

# 4 Application

To demonstrate our methods, we apply them to the setting studied in Section 5.1 of Horowitz (2011). Horowitz (2011) estimates a shape-invariant Engel curve for

food using data from the British Family Expenditure Survey.[4] Horowitz's application is in turn based on Blundell et al. (2007) who also carry out NPIV estimation of shape-invariant Engel curves and use the same data.

**Endogeneity.** In observational settings, the share of household wealth allocated to nondurables is decided by the household. Therefore, the budget for nondurables depends on household preferences. The same preferences also influence the allocation of the nondurable budget across categories such as food.

To tackle this problem Blundell et al. (2007) and Horowitz (2011) use household income as an instrument for nondurables expenditure. There are many reasons to think that household income and preferences are correlated. Both tastes and income may depend on household size and socio-economic status, and those with expensive tastes may seek high-paying jobs. However, some shocks to household income may result from exogenous external factors like shocks to production costs in an employed householder's firm. If one controls for a sufficiently rich set of household covariates this may absorb the endogenous variation in income leaving only the exogenous variation. If the data do not contain rich enough household observables, then some endogeneity likely remains and the income instrument is unlikely to be fully valid.

Blundell et al. (2007) and Horowitz (2011) can only control for a limited set of demographic variables.[5] Therefore, in this setting instrumental validity may not hold exactly and it is of interest to assess what empirical findings are robust to some failure of instrumental validity.

**Data and variables.** The sample consists of $n = 1655$ households drawn from the British Family Expenditure Survey. Following Horowitz (2011), we restrict attention to married couples with an employed head-of-household aged 25–55. Let $Y_i$ be the share of non-durable expenditure spent on food, let $X_i$ be the logarithm of total non-durable expenditure, and let $Z_i$ be the logarithm of household income. Our object of interest is the function $h_0 : S_X \to [0, 1]$, where $S_X$ denotes the support of $X$.

[4] We made use of the data file that accompanies Horowitz (2011) and adapted the accompanying code in order to evaluate Horowitz's estimator and B-spline bases for our own methods.

[5] Both papers control for demographics by analyzing a homogeneous sub-sample. Blundell et al. (2007) incorporate additional dummy variable controls.

**Shape restrictions and function class.** We impose bounded curvature and range restrictions by taking

$$\mathcal{H}(c) = \Big\{ h \in C^2(S_X) \ : \ 0 \leq h \leq 1, \ \sup_{x \in S_X} |h''(x)| \leq c \Big\}.$$

We report results for $c = 1.0$, $c = 1.25$ and $c = 1.5$ representing tighter, moderate, and looser curvature constraints. For comparison, the magnitude of the second derivatives of Horowitz's estimated structural function never exceed 0.5.

In the computation, $X$ is normalized to the unit interval before constructing the spline basis. The curvature bounds reported are stated with respect to the original log-expenditure scale.[6]

**Calibrating departures from instrument validity.** Following the calibration approach in Section 2.1, we index the magnitude of misspecification by a scalar $b$. In particular, for a given $\tau \in (0, 1/2)$, we set

$$\hat{b}(\tau) = \hat{q}_\eta(0.5 + \tau) - \hat{q}_\eta(0.5 - \tau),$$

where $\hat{q}_\eta(\cdot)$ denotes the empirical quantile function of residuals $\hat{\eta}_i = Y_i - \hat{g}_n(Z_i)$. Here $\hat{g}_n$ is a nonparametric estimate of the reduced-form function $g_0(z) = \mathbb{E}[Y \mid Z = z]$ (implemented via a spline series regression in $Z$). We consider $\tau \in \{0.05, 0.1, 0.15\}$, which we interpret as mild, moderate, and severe deviations from exact instrument validity.[7]

**Sieve approximation and cross-validation tuning.** Horowitz (2011) uses cubic B-spline bases with evenly spaced knots for the structural sieve $\Phi_n$.[8] We also use spline bases, but allow the first-stage and second-stage basis dimensions to differ. For the reported finite-sample calculation, we set the dimension of $\Phi_n$ to $K_n = 10$. For the first stage, we choose the dimension $L_n$ of the $Z$-basis jointly with a ridge (Tikhonov) regularization parameter by cross-validation. Specifically, we consider a

[6] Let $x^{\text{norm}} = \frac{x^{\text{orig}} - \min(x^{\text{orig}})}{\Delta_x}$ be the normalized $X$ with $\Delta_x = \max(x^{\text{orig}}) - \min(x^{\text{orig}})$. A curvature bound $c$ on the original log-expenditure scale is implemented as the bound $c\Delta_x^2$.

[7] In the implementation, the implied values of $\hat{b}(\tau)$ are computed directly from the sample residuals $\hat{\eta}_i$ using the definition above.

[8] See de Boor (2014) for a practical introduction to B-splines.

grid of candidate spline dimensions $L \in \mathcal{L}$ and ridge penalties $\lambda \in \Lambda$. We then use the one-standard-error cross-validation rule described in Appendix B to select $(L, \lambda)$.

**Grids and implementation choices.** Taking $S_H = S_X$, we evaluate the pointwise identified intervals for $h_0(x)$ on a grid $\widetilde{S}_H$ of 80 equally spaced points between the minimum and maximum observed values of $X$. We construct $\widetilde{S}_Z$ as 80 equally spaced points between the 0.005 and 0.995 empirical quantiles of $Z$. These are the constraints used for the reported computation. Trimming the instrument grid does not by itself approximate a moment restriction on the full support; applying the full-support theory requires control of the omitted tails or an explicitly restricted instrument domain.

**Sampling uncertainty.** The reported calculation uses subsamples of size $m = \lfloor n^{0.75} \rfloor$, drawn without replacement, with $B = 1000$ replications. It holds the calibrated $\hat{b}(\tau)$ fixed across subsamples. We describe the calculation explicitly, while distinguishing it from an established coverage result.

Subsampling inference requires a suitable normalizing sequence $\rho_n \to \infty$ and a limiting distribution for the normalized estimation error, together with $m \to \infty$, $m/n \to 0$, $\rho_m/\rho_n \to 0$, negligible Monte Carlo error, and the corresponding bias and quantile-continuity conditions (Politis and Romano, 1994). A convergence-rate bound alone does not supply these conditions. In a growing nonparametric sieve, $\rho_n$ is generally not $\sqrt{n}$. Neither the needed limit law nor negligibility of sieve, grid, ridge, and calibration errors at the inference scale is established here. Holding $\hat{b}(\tau)$ fixed also does not automatically account for its sampling variation.

The formulas below use a generic $\rho_n$; the supplied figure used $\rho_n = \sqrt{n}$. Its shaded regions should therefore be interpreted as illustrative sampling-uncertainty summaries, with nominal coverage requiring separate justification for that implementation.

The pointwise identified intervals have lower and upper envelope functions

$$\underline{\theta}(x) = \inf_{h \in \Theta_b} h(x), \qquad \bar{\theta}(x) = \sup_{h \in \Theta_b} h(x).$$

We first use subsampling to approximate the sampling uncertainty in the estimated

envelopes $\hat{\underline{\theta}}(x)$ and $\hat{\bar{\theta}}(x)$. For subsample $s \in \{1, \ldots, B\}$, define

$$T_{\ell,s} = \sup_{x \in \widetilde{S}_H} \rho_m \left| \hat{\underline{\theta}}^{(s)}(x) - \hat{\underline{\theta}}(x) \right|,$$

and

$$T_{u,s} = \sup_{x \in \widetilde{S}_H} \rho_m \left| \hat{\bar{\theta}}^{(s)}(x) - \hat{\bar{\theta}}(x) \right|.$$

Let $q_{\ell,1-\alpha/2}$ and $q_{u,1-\alpha/2}$ denote the empirical $(1-\alpha/2)$-quantiles of $\{T_{\ell,s}\}_{s=1}^{B}$ and $\{T_{u,s}\}_{s=1}^{B}$, respectively.

The corresponding Bonferroni-adjusted candidate bands for the two envelope functions are:

$$\widehat{\mathcal{C}}_\ell(x) = \left[ \hat{\underline{\theta}}(x) - \frac{q_{\ell,1-\alpha/2}}{\rho_n},\ \hat{\underline{\theta}}(x) + \frac{q_{\ell,1-\alpha/2}}{\rho_n} \right],$$

and

$$\widehat{\mathcal{C}}_u(x) = \left[ \hat{\bar{\theta}}(x) - \frac{q_{u,1-\alpha/2}}{\rho_n},\ \hat{\bar{\theta}}(x) + \frac{q_{u,1-\alpha/2}}{\rho_n} \right], \qquad x \in \widetilde{S}_H.$$

Under the distributional and bias conditions just described, these would provide simultaneous coverage of the two envelopes over the evaluation grid.

Because the maintained assumptions imply

$$\underline{\theta}(x) \leq h_0(x) \leq \bar{\theta}(x), \qquad x \in S_X,$$

simultaneously valid envelope bands would imply an outer band for the structural function:

$$\widehat{\mathcal{C}}_h(x) = \left[ \max\left\{0, \hat{\underline{\theta}}(x) - \frac{q_{\ell,1-\alpha/2}}{\rho_n}\right\},\ \min\left\{1, \hat{\bar{\theta}}(x) + \frac{q_{u,1-\alpha/2}}{\rho_n}\right\} \right], \qquad x \in \widetilde{S}_H.$$

The truncation by 0 and 1 uses the maintained range restriction $h_0(x) \in [0,1]$. If both envelope bands have their stated asymptotic coverage, Bonferroni gives coverage at least $1-\alpha$ for this outer band over the evaluation grid. The present rate results do not establish that premise for the reported calculation.

We also report the midpoint estimator

$$\hat{\theta}^{\text{mid}}(x) = \frac{\hat{\underline{\theta}}(x) + \hat{\bar{\theta}}(x)}{2},$$

which provides the pointwise minimax summary described in Section 1.3. This does not assert minimaxity under integrated squared loss over the joint identified set.

**Results.** Figure 4.1 reports the estimated identified sets for $h_0(x)$ across combinations of the misspecification calibration parameter $\tau$ and the curvature bound $c$. Columns correspond to increasing values of $\tau \in \{0.05, 0.10, 0.15\}$, while rows correspond to increasing values of $c \in \{1.0, 1.25, 1.5\}$. In each panel, the red and blue solid curves are the estimated lower and upper envelopes, $\hat{\underline{\theta}}(x)$ and $\hat{\bar{\theta}}(x)$, respectively. The red and blue shaded regions are Bonferroni-adjusted subsampling uncertainty bands for these two envelope functions. The black dash-dotted curves give the implied outer uncertainty band for the structural function $h_0(x)$. The green solid curve is the midpoint estimator $\hat{\theta}^{\text{mid}}(x)$, and the gray dashed curve reproduces the NPIV point estimate reported by Horowitz (2011).

The figure illustrates the robustness–smoothness trade-off emphasized in the theory. Holding $c$ fixed, moving from left to right increases $\tau$, and therefore increases the calibrated bound $b(\tau)$ on deviations from the exact NPIV moment condition. This weakens the maintained instrument-validity restriction and enlarges the estimated identified set. Holding $\tau$ fixed, moving from top to bottom increases $c$, thereby weakening the curvature restriction and expanding the class of admissible structural functions. Thus the width of the estimated set reflects both the allowed degree of instrument invalidity and the strength of the maintained smoothness restriction.

The displayed point estimates are consistent with a declining Engel curve over much of the expenditure range. However, the more detailed nonmonotonic feature at low expenditure levels emphasized by Horowitz (2011) is sensitive to modest departures from exact instrument validity. Overlap of pointwise envelopes alone does not establish that every admissible function is monotone; that conclusion requires optimizing the relevant differences or imposing monotonicity jointly. In this sense, the application illustrates the value of reporting both the identified set and the midpoint estimator: the envelopes summarize pointwise ranges under the sensitivity analysis, while the midpoint provides a stable point summary of the partially identified structural relationship.

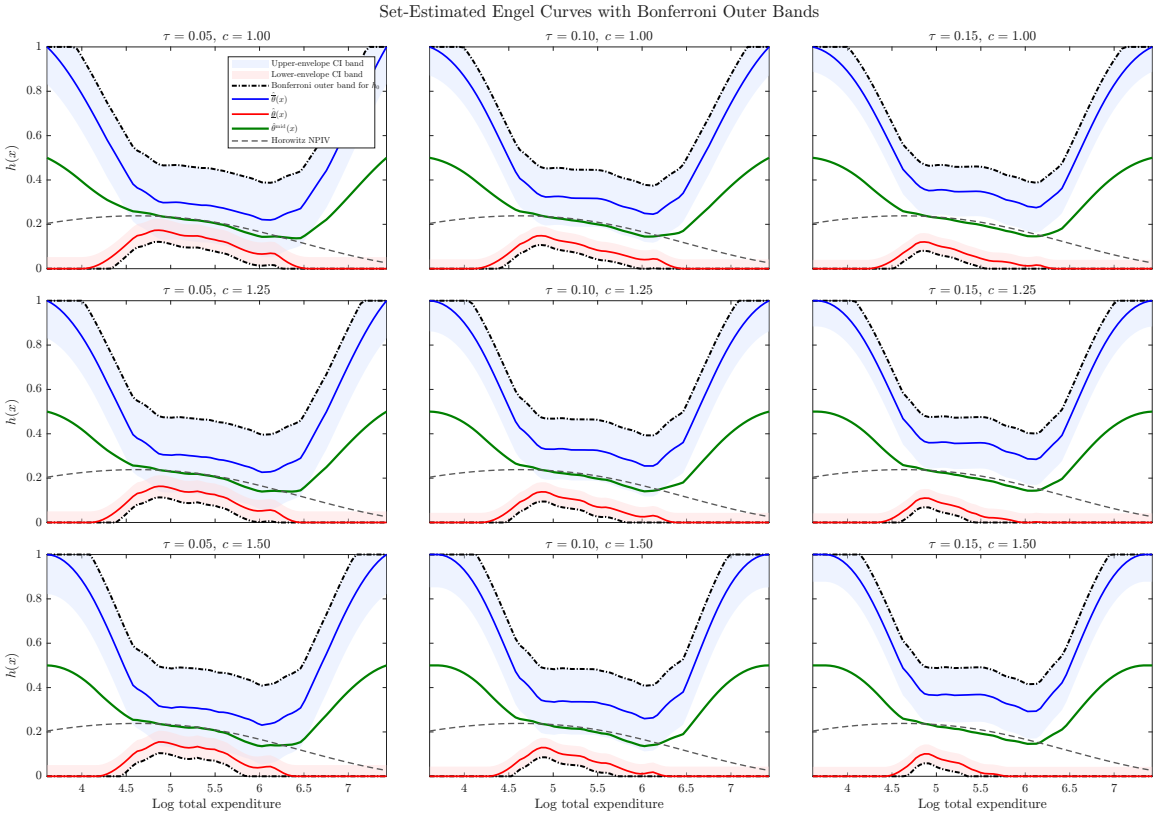



Figure 4.1: Set-estimated Engel curves across misspecification levels and curvature restrictions.

Notes: Columns correspond to $\tau \in \{0.05, 0.10, 0.15\}$, and rows correspond to $c \in \{1.0, 1.25, 1.5\}$. The red and blue solid curves are the estimated lower and upper envelopes of the identified set for $h_0(x)$. The red and blue shaded regions are Bonferroni-adjusted subsampling uncertainty bands for the lower and upper envelope functions. The black dash-dotted curves are the implied outer uncertainty band for $h_0(x)$, obtained from the lower edge of the lower-envelope band and the upper edge of the upper-envelope band. The green solid curve is the midpoint estimator $\hat{\theta}^{\text{mid}}(x)$. The gray dashed curve is the NPIV estimate from Horowitz (2011). The first-stage reduced-form regressions use the cross-validated tuning parameters $L_n = 4$ and $\lambda = 2.85 \times 10^{-9}$. Subsampling uses $m = \lfloor n^{0.75} \rfloor$ and $B = 1000$, with the reported root-$n$ normalization. Nominal coverage of these illustrative bands is not established by the rate theory.

# 5 Misspecification Diagnostics and the Falsification Frontier

The sensitivity analysis developed above treats the discrepancy bound $b$ as a user-specified measure of tolerated misspecification. A complementary question is whether the data themselves imply that some positive discrepancy is required for the maintained NPIV model to be feasible. This is closely related in spirit to the misspecification diagnostics developed for moment-inequality models by Andrews and Kwon (2024). In our setting, the natural diagnostic is the minimum discrepancy level needed for the set of admissible structural functions to be nonempty.

## 5.1 Population falsification frontier

Fix a shape-restricted class $\mathcal{H}(c)$. For a given discrepancy level $b \geq 0$, recall the feasible set

$$\Theta_b(g_0; c) = \{h \in \mathcal{H}(c) : \ \|g_0 - \mathbb{A}[h]\|_\infty \leq b\}.$$

We define the *minimum required discrepancy* (or *falsification frontier*) at smoothness level $c$ by

$$b^\star(c) = \inf_{h \in \mathcal{H}(c)} \|g_0 - \mathbb{A}[h]\|_\infty. \tag{5.1}$$

Equivalently,

$$b^\star(c) = \inf\{b \geq 0 : \Theta_b(g_0; c) \neq \emptyset\}.$$

For every $b > b^\star(c)$ the feasible set is nonempty, and for $b < b^\star(c)$ it is empty. At $b = b^\star(c)$, feasibility additionally requires attainment of the infimum, for example when $\mathcal{H}(c)$ is compact and the residual norm is continuous.

For a fixed benchmark $b_0 \geq 0$, a natural null hypothesis is

$$H_0(b_0, c) : \qquad b^\star(c) \leq b_0, \tag{5.2}$$

against the alternative $H_1(b_0, c) : b^\star(c) > b_0$. When the infimum is attained, the null states that the model is feasible under discrepancy level $b_0$; in general it states that its distance from feasibility is no greater than $b_0$. The alternative states that no structural function in $\mathcal{H}(c)$ can rationalize the data unless the discrepancy bound is taken larger than $b_0$. Because the classes $\mathcal{H}(c)$ are nested, the falsification frontier is

monotone in $c$.

**Proposition 5.1.** *Suppose $c_1 \leq c_2$ implies $\mathcal{H}(c_1) \subseteq \mathcal{H}(c_2)$. Then*

$$b^\star(c_2) \leq b^\star(c_1).$$

Proposition 5.1 formalizes the natural idea that weaker smoothness restrictions make the model easier to fit and therefore cannot raise the minimum discrepancy required for feasibility.

## 5.2 Sample analog

Given first-stage estimators $\hat{g}_n$ and $\hat{\Pi}_n$, a sieve basis $\Phi_n$, and grids $\widetilde{S}_H$ and $\widetilde{S}_Z$, define the sample analog of $b^\star(c)$ by

$$\begin{aligned} \hat{b}_n^\star(c) = \min_{t \in \mathbb{R},\, \beta \in \mathbb{R}^{K_n}} \; & t \\ \text{s.t.} \quad & |\hat{g}_n(z) - \hat{\Pi}_n(z)'\beta| \leq t, \qquad \forall z \in \widetilde{S}_Z, \\ & \mathbb{T}[\Phi_n'](x)\beta \leq C_c(x), \qquad \forall x \in \widetilde{S}_H, \end{aligned} \tag{5.3}$$

where $C_c$ denotes the bound vector defining $\mathcal{H}(c)$. This is a linear programming problem: the scalar $t$ is the maximal discrepancy needed to make the sample analog of the model feasible.

The estimator $\hat{b}_n^\star(c)$ is useful both as a diagnostic and as a summary statistic. For each $c$, it estimates the smallest discrepancy level consistent with the data. Plotting $c \mapsto \hat{b}_n^\star(c)$ yields an estimated falsification frontier that summarizes how much misspecification is needed to reconcile the model with the data at different levels of smoothness.

## 5.3 Consistency and a simple misspecification test

Let

$$r_n = a_n + \kappa_n + \iota_n \xi_n D_{H,n} + \iota_n G_n D_{Z,n}.$$

The next theorem shows that the estimated falsification frontier converges at the same rate as the set estimator. A direct proof controls the residual objective without treating the unbounded discrepancy variable as part of a bounded function class.

**Theorem 5.2.** *Fix $c$. Suppose Assumptions 3.1–3.3 hold for $\mathcal{H}(c)$. For some fixed $B > 1 + \|g_0 - \mathbb{A}[h^s]\|_\infty$, suppose $\sup_{h \in \Theta_B} \|h\|_\infty < \infty$. Then $|\hat{b}_n^\star(c) - b^\star(c)| = O_p(r_n) = o_p(1)$.*

Theorem 5.2 immediately yields a simple consistent test of misspecification at a benchmark discrepancy level $b_0$.

**Corollary 5.3.** *Fix $c$ and $b_0 \geq 0$. Let $\lambda_n \downarrow 0$ be any deterministic sequence satisfying*

$$r_n = o(\lambda_n).$$

*Consider the test*

$$\varphi_n(b_0, c) = \mathbf{1}\{\hat{b}_n^\star(c) > b_0 + \lambda_n\}.$$

*Then:*

1. *under $H_0(b_0, c)$, $P(\varphi_n(b_0, c) = 1) \to 0$;*
2. *if $b^\star(c) > b_0$, then $P(\varphi_n(b_0, c) = 1) \to 1$.*

Corollary 5.3 provides a simple pointwise consistent falsification test. It is conservative under the null because of the buffer $\lambda_n$, but consistent against any fixed alternative for which the true minimum required discrepancy exceeds the benchmark $b_0$. A test with a nonzero asymptotic significance level would require additional distributional analysis.

The diagnostic developed here is related in spirit to the misspecification diagnostics of Andrews and Kwon (2024), but it differs in both object and structure. Andrews and Kwon study general moment inequality models and develop diagnostics and inference procedures that remain informative when misspecification renders the identified set empty. By contrast, our setting is a nonparametric conditional moment model with an ill-posed inverse structure, and our diagnostic object is the minimum discrepancy level $b^\star(c)$ required for feasibility within a smoothness class $\mathcal{H}(c)$. Thus our falsification frontier is tailored to NPIV and explicitly incorporates the trade-off between misspecification tolerance and functional regularization.

# References


Ai, C. and Chen, X. (2003). Efficient estimation of models with conditional moment restrictions containing unknown functions. *Econometrica*, 71(6):1795–1843.

Altonji, J., Elder, T., and Taber, C. (2005). Selection on observed and unobserved variables: Assessing the effectiveness of catholic schools. *Journal of Political Economy*, 113:151–184.

Andrews, D. W. and Kwon, S. (2024). Misspecified moment inequality models: Inference and diagnostics. *Review of Economic Studies*, 91(1):45–76.

Andrews, I., Gentzkow, M., and Shapiro, J. M. (2017). Measuring the sensitivity of parameter estimates to estimation moments. *The Quarterly Journal of Economics*, 132:1553–1592.

Angrist, J., Imbens, G., and Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *Journal of the American Statistical Association*.

Armstrong, T. B. and Kolesar, M. (2018). Sensitivity analysis using approximate moment condition models. *Cowles Foundation Discussion Papers*.

Belloni, A., Chernozhukov, V., Chetverikov, D., and Kato, K. (2015). Some new asymptotic theory for least squares series: Pointwise and uniform results. *Journal of Econometrics*, 186:345–366.

Blundell, R., Chen, X., and Kristensen, D. (2007). Semi-nonparametric IV estimation of shape-invariant Engel curves. *Econometrica. Journal of the Econometric Society*, 75(6):1613–1669.

Bonhomme, S. and Weidner, M. (2022). Minimizing sensitivity to model misspecification. *Quantitative Economics*, 13(3):907–954.

Bound, J., Jaeger, D. A., and Baker, R. M. (1995). Problems with instrumental variables estimation when the correlation between the instruments and the endogeneous explanatory variable is weak. *Journal of the American Statistical Association*, 90:443.

Chen, X. and Pouzo, D. (2012). Estimation of nonparametric conditional moment models with possibly nonsmooth generalized residuals. *Econometrica*.

Conley, T. G., Hansen, C. B., McCulloch, R. E., and Rossi, P. E. (2008). A semi-parametric Bayesian approach to the instrumental variable problem. *Journal of Econometrics*, 144(1):276–305.

de Boor, C. (2014). (b)asic-spline basics.

Deaner, B. (2025). The trade-off between flexibility and robustness in instrumental variables analysis. *American Economic Review*, 12.

Dominitz, J. and F. Manski, C. (2017). More data or better data? a statistical decision problem. *The Review of Economic Studies*, 84(4):1583–1605.

Dudley, R. M. (1967). The sizes of compact subsets of hilbert space and continuity of gaussian processes. *Journal of Functional Analysis*, 1:290–330.

Horowitz, J. L. (2011). Applied nonparametric instrumental variables estimation. *Econometrica. Journal of the Econometric Society*, 79(2):347–394.

Imbens, G. W. (2003). Sensitivity to exogeneity assumptions in program evaluation. *American Economics Review, P&P*, 93:126–132.

Kankanala, S. (2025). Generalized bayes in conditional moment restriction models.

Masten, M. A. and Poirier, A. (2018). Identification of treatment effects under conditional partial independence. *Econometrica*.

Newey, W. K. and Powell, J. L. (2003). Instrumental variable estimation of nonparametric models. *Econometrica*, 71:1565–1578.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37:187–204.

Politis, D. N. and Romano, J. P. (1994). Large sample confidence regions based on subsamples under minimal assumptions. *The Annals of Statistics*, pages 2031–2050.

Rudelson, M. (1999). Random vectors in the isotropic position. *Journal of Functional Analysis*, 164:60–72.

Song, K. (2014). Point decisions for interval–identified parameters. *Econometric Theory*, 30(2):334–356.

# A Extensions to the set estimator

For a fixed number $J$ of instrument vectors $Z_j$, replace the single residual bound with known lower and upper functions:

$$b_{1,j}(Z_j) \le E[Y - h(X) \mid Z_j] \le b_{2,j}(Z_j), \qquad j = 1, \dots, J.$$

The original restriction is the case $J = 1$, $b_{1,1} = -b$ and $b_{2,1} = b$. Define $g_{0,j}(z) = E[Y \mid Z_j = z]$ and $\Pi_{n,j}(z) = E[\Phi_n(X) \mid Z_j = z]$. The feasible constraints are

$$b_{1,j}(z) \le \hat{g}_{n,j}(z) - \hat{\Pi}_{n,j}(z)'\beta \le b_{2,j}(z), \qquad z \in \widetilde{S}_{Z_j}, \quad j \le J,$$

together with the shape restrictions in (2.2). Theorem C.1 gives the corresponding rate when there is a fixed positive feasibility margin.

# B First-stage cross-validation

The accompanying cross-validation specification uses a common ridge series fit for the response vector $(Y_i, \Phi_n(X_i)')'$. Candidate pairs consist of a first-stage dimension $L$ and a ridge penalty $\lambda$. For each pair, compute mean held-out squared prediction error across folds, optionally scaling the response components to comparable units using training-fold scales. Let $\widehat{\mathrm{CV}}_{\min}$ be the smallest mean score and let $\widehat{\mathrm{SE}}_{\min}$ be the standard deviation of its fold scores divided by the square root of the number of folds. The one-standard-error rule retains pairs whose score is at most $\widehat{\mathrm{CV}}_{\min} + \widehat{\mathrm{SE}}_{\min}$, then chooses the smallest retained $L$ and, among ties, the largest $\lambda$. This is a tuning heuristic, not a coverage or consistency theorem. For reproducibility, the candidate grid, fold construction, scaling, and selected tuning parameters should accompany the empirical code.

# C Proofs

**Proof of Proposition 3.1.** Put $d = b - \varepsilon_0 > \eta$ and $c^* = \mathbb{L}[h^*]$. Both feasible sets contain $h^*$. If $\tilde{h}$ is perturbed-feasible, its population residual norm is at most $b + \eta$.

Hence

$$h^\circ = \frac{d}{d+\eta}\tilde{h} + \frac{\eta}{d+\eta}h^*$$

is population-feasible, by convexity and the residual bound $\{d(b+\eta)+\eta\varepsilon_0\}/(d+\eta) = b$. It follows for every such $\tilde{h}$ that

$$\mathbb{L}[\tilde{h}] - \bar{\theta}_{\mathbb{L}}(b) \leq \frac{\eta}{d}\{\bar{\theta}_{\mathbb{L}}(b) - c^*\}.$$

Applying the same argument to $-\mathbb{L}$ proves finiteness of both perturbed endpoints as well as the outward perturbation bounds. Conversely, for each population-feasible $h$, the function $(1-\eta/d)h + (\eta/d)h^*$ is perturbed-feasible: its population residual is at most $b-\eta$, and its perturbation error is at most $\eta$. Taking suprema gives

$$\bar{\theta}_{\mathbb{L}}(b) - \widetilde{\bar{\theta}}_{\mathbb{L}}(b) \leq \frac{\eta}{d}\{\bar{\theta}_{\mathbb{L}}(b) - c^*\}.$$

Replace $\mathbb{L}$ by $-\mathbb{L}$ to obtain the lower-endpoint result. The midpoint bound follows by adding the two endpoint bounds and dividing by two. No endpoint optimizer need exist. ∎

**Proof of Theorem 3.2.** Throughout the proof, constants are independent of $n$, $b$, and $\mathbb{L}$ in the specified ranges. We work on events whose probabilities tend to one, on which the Lipschitz bounds hold and the small-error inequalities below hold simultaneously for all $b \in [\underline{b}_n, \bar{b}]$.

*A common feasible anchor.* Choose $q_n^p = q_n[h^p]$ from Assumption 3.2. Set

$$D = 1 + \varepsilon_0 + \|g_0 - \mathbb{A}[h^s]\|_\infty, \quad t_b = \min\{1/2, (b-\varepsilon_0)/(2D)\}, \quad q_{n,b} = (1-t_b)q_n^p + t_b q_n^s.$$

These functions have norms bounded by a fixed constant $A_0$. Their shape slack is at least $t_b\sigma$, and

$$\|g_0 - \mathbb{A}[q_{n,b}]\|_\infty \leq \varepsilon_0 + t_b D + \kappa_n \leq b - (b-\varepsilon_0)/2 + \kappa_n.$$

There is $c_0 > 0$ such that both the shape slack and the population moment slack before subtracting $\kappa_n$ are at least $c_0 s(b)$. Since $q_{n,b} \in \mathcal{H}_n$, its empirical residual differs from its population residual by at most $\eta_n$. On the event $\kappa_n + \eta_n \leq c_0 s(\underline{b}_n)/2$, the function $q_{n,b}$ therefore satisfies every population and continuum empirical constraint

with slack at least

$$\mu_b = c_0 s(b)/2.$$

This event has probability tending to one. In particular, all the feasible sets used below are nonempty.

We repeatedly use the following elementary repair. If a candidate violates each affine inequality by at most $e \geq 0$, and an anchor has slack $\mu > 0$ in each inequality, their convex combination with weight $e/(\mu + e)$ on the anchor satisfies all the inequalities. Its distance from the candidate is at most $e\|h - q\|_\infty/(\mu + e)$.

*Population sieve approximation.* Let $F_b^* = \{h \in \mathcal{H}_n : \|g_0 - \mathbb{A}[h]\|_\infty \leq b\}$. For $h \in \Theta_b$, the feasible approximation $q_n[h]$ satisfies the moment bound with violation at most $\kappa_n$, since $\|\mathbb{A}\|_{\text{op}} \leq 1$. Repair it toward $q_{n,b}$. Both $\|h\|_\infty \leq M$ and $\|q_n[h]\|_\infty \leq M + \kappa_n$, so the resulting element of $F_b^*$ is within $C\kappa_n/s(b)$ of $h$. Since $F_b^* \subseteq \Theta_b$, upper and lower support values differ by at most $CC_L\kappa_n/s(b)$.

*First-stage perturbation.* Let

$$F_b^\circ = \{h = \Phi_n'\beta \in \mathcal{H}_n : \|\hat{g}_n - \hat{\Pi}_n'\beta\|_\infty \leq b\}.$$

For any $h \in F_b^\circ$, the population moment violation is at most $\eta_n$. Repair toward $q_{n,b}$ to obtain an element of $F_b^*$. Solving the convex-combination identity for $h$ gives

$$\|h\|_\infty \leq (1 + \eta_n/\mu_b)M + (\eta_n/\mu_b)A_0.$$

Because $\eta_n/s(\underline{b}_n) = o_p(1)$, this is bounded by a fixed constant $M_1$ simultaneously in $b$. The same repair in the opposite direction maps $F_b^*$ into $F_b^\circ$. The distances in both directions are at most $C\eta_n/s(b)$, so the corresponding endpoint differences are at most $CC_L\eta_n/s(b)$.

*Finite grids, including boundedness.* Let $F_b^g$ denote the actual grid-feasible sieve set. If $h = \Phi_n'\beta \in F_b^g$, the Lipschitz assumptions imply that its maximum violation of any continuum empirical inequality is at most

$$e_n(h) \leq \{\xi_n D_{H,n} + G_n D_{Z,n}\}(1 + \|\beta\|_2) \leq d_n(1 + \|h\|_\infty).$$

This step does not apply a first-stage bound to a function outside $\mathcal{H}$. Repairing all the continuum empirical inequalities toward $q_{n,b}$ yields an element of $F_b^\circ$, whose norm

is at most $M_1$. Solving again for $h$ gives

$$\|h\|_\infty \le M_1 + \frac{d_n(M_1 + A_0)}{\mu_b}(1 + \|h\|_\infty).$$

Uniformly in $b$, the coefficient of $1 + \|h\|_\infty$ is at most $1/2$ for all sufficiently large $n$, because $d_n/s(\underline{b}_n) \to 0$. Thus $\|h\|_\infty \le 2M_1 + 1$ for every grid-feasible $h$. In particular, the grid programs are bounded, and $e_n(h) \le (2M_1 + 2)d_n$. Their repaired functions are at distance at most $Cd_n/s(b)$. Since $F_b^\circ \subseteq F_b^g$, the grid endpoint differences are at most $CC_L d_n/s(b)$.

Combining the three comparisons shows, on the same event,

$$\sup_{b\in[\underline{b}_n,\bar{b}]} s(b) \sup_{\|\mathbb{L}\|_{\mathrm{op}}\le C_L} \max_{\diamond\in\{\mathrm{lower,upper}\}} |\hat{\theta}_{\mathbb{L}}^{\diamond}(b) - \theta_{\mathbb{L}}^{\diamond}(b)| \le CC_L(\kappa_n + \eta_n + d_n).$$

Here the superscripts in the last display denote the corresponding lower and upper endpoints. Substituting $\eta_n = O_p(a_n)$ proves (3.3). The average of the two endpoint errors bounds the midpoint error. Point evaluation has norm one. The programs may be assigned any finite value on the exceptional event of infeasibility; its probability tends to zero uniformly over the stated range. ∎

**Proof of Corollary 3.3.** If $\omega(t)$ did not tend to zero, there would be $\delta > 0$, $t_j \downarrow 0$, and $h_j \in \overline{\mathcal{H}}$ with $\|h_j - h_0\|_\infty \ge \delta$ and $\|\mathbb{A}[h_j - h_0]\|_\infty \le t_j$. Compactness supplies a uniformly convergent subsequence with limit $h_\infty \in \overline{\mathcal{H}}$. Continuity of $\mathbb{A}$ gives $\mathbb{A}[h_\infty] = g_0$, contradicting uniqueness and the distance bound. Thus $\omega(t) \to 0$. Every $h \in \Theta_b$ satisfies $\|h - h_0\|_\infty \le \omega(b)$. Its values, and therefore both pointwise endpoints and their midpoint, lie within $\omega(b)$ of $h_0$. For a bounded linear functional the same reasoning gives radius $\|\mathbb{L}\|_{\mathrm{op}}\omega(b)$ around $\mathbb{L}[h_0]$. Apply Theorem 3.2 with $\underline{b}_n = b_n$ and use the triangle inequality. The power rate follows by balancing $b_n^\alpha$ and $r_n/b_n$; this choice satisfies $r_n = o(b_n)$. ∎

**Theorem C.1** (General conditional-moment bands)**.** *Consider the extension in Appendix A with a fixed number $J$ of instrument vectors. Suppose Assumption 3.2 holds, the population feasible set is bounded in supremum norm, and some $h^p \in \mathcal{H}$ satisfies every moment band with a common positive slack $\delta$. Suppose the first-stage error*

*bound holds for every $j \le J$ with rate $a_n$, and the grid conditions hold with*

$$D_{Z,n} = \max_{j \le J} \sup_{z \in S_{Z_j}} \min_{z' \in \widetilde{S}_{Z_j}} \|z - z'\|_2,$$

*where the two moment-bound functions and both first-stage functions have Lipschitz constants at most $G_n$. Then both endpoint errors are $O_p(r_n)$, uniformly over bounded linear functionals with norm at most $C_L$.*

**Proof.** The moment-bound functions are bounded on the bounded supports under the stated Lipschitz conditions, and the residual of $h^s$ is bounded. A sufficiently small fixed mixture of $h^p$ and $h^s$ therefore retains positive moment slack. Its sieve version, formed from $q_n[h^p]$ and $q_n^s$, has positive shape slack and, eventually, positive population and empirical moment slack. Take the minimum of these finitely many margins as a fixed $\mu > 0$. The three repair comparisons in the preceding theorem now apply with this constant margin. The maximum first-stage error over the fixed $J$ is $O_p(a_n)$. Differences between the Lipschitz moment bound functions at grid and off-grid points add at most a fixed multiple of $G_n D_{Z,n}$ to the grid error. The same norm inequality establishes boundedness of every grid-feasible function. Adding the three endpoint comparisons proves the result. ∎

**Lemma C.2** (Uniform series first stage)**.** *Under Assumptions 3.4 and 3.5, put $\hat{\gamma}[h] = \hat{Q}_n^{-1} n^{-1} \sum_i \Psi_n(Z_i) h(X_i)$ on the nonsingular event. Then*

$$\sup_{h \in \mathcal{H}} \|\Psi_n' \hat{\gamma}[h] - \mathbb{A}[h]\|_\infty = O_p\left( \frac{\bar{\xi}_n}{\sqrt{n}} \{1 + \sqrt{\log(2\ell_n)} + \sqrt{L_n} R_n(s)\} + R_n(s) \right).$$

*This bound is $o_p(1)$ whenever its displayed deterministic rate tends to zero.*

**Proof.** The entropy condition makes $\mathcal{H}$ totally bounded. In particular,

$$M_H := \sup_{h \in \mathcal{H}} \|h\|_\infty < \infty.$$

Indeed, finiteness of the integral ensures a finite cover at some positive radius, which already gives boundedness; monotonicity then ensures finite covers at every positive radius. Write $C_A = \int \ell \, d\mu_X$. Differentiating under the integral, and integrating the Hölder difference bounds, gives

$$\|\mathbb{A}[h]\|_{C^s} \le C_A \|h\|_\infty.$$

This also holds for differences of two functions in $\mathcal{H}$.

Whiten $\Psi_n$ by $Q_n^{-1/2}$. Uniform eigenvalue bounds change the bounds on its norm and its normalized-direction Lipschitz constant only by fixed factors, and do not change the series projection or fitted values. We can therefore suppose $Q_n = I$. Let $\gamma[h] = E[\Psi_n(Z)h(X)]$, and define

$$e_i[h] = h(X_i) - \mathbb{A}[h](Z_i), \qquad v_h(z) = \mathbb{A}[h](z) - \Psi_n(z)'\gamma[h].$$

Then $|e_i[h]| \le 2M_H$, $|e_i[h] - e_i[k]| \le 2\|h-k\|_\infty$, and

$$\|v_h\|_\infty \le C_A M_H R_n(s), \qquad \|v_h - v_k\|_\infty \le C_A R_n(s)\|h-k\|_\infty. \tag{C.1}$$

The matrix law of large numbers (Rudelson, 1999; Belloni et al., 2015, Lemma 6.2) gives $\|\hat{Q}_n - I\|_{\mathrm{op}} = o_p(1)$. In what follows we restrict to the event where its eigenvalues lie in $[1/2, 2]$. All constants in the conditional bounds on this event are deterministic.

The least squares decomposition is

$$\begin{aligned}\sqrt{n}\,\alpha_n(z)'(\hat{\gamma}[h] - \gamma[h]) = {} & \alpha_n(z)'\hat{Q}_n^{-1} n^{-1/2} \sum_i \Psi_n(Z_i) e_i[h] \\ & + \alpha_n(z)'\hat{Q}_n^{-1} n^{-1/2} \sum_i \Psi_n(Z_i) v_h(Z_i).\end{aligned}$$

For the first process, condition on $Z_1, \ldots, Z_n$ and symmetrize the independent, conditionally mean-zero errors using independent Rademacher variables $\epsilon_i$. The empirical $L^2$ distance between the coefficient arrays indexed by $(z, h)$ and $(z', k)$ is bounded by

$$C\{M_H \ell_n \|z - z'\|_2 + \|h - k\|_\infty\}.$$

Split the difference into its $z$ and $h$ components and use

$$n^{-1} \sum_i \Psi_n(Z_i)\Psi_n(Z_i)' = \hat{Q}_n$$

to obtain this metric bound. Adjoin the zero array and the negatives of the arrays when bounding the absolute supremum; this adds only a constant to the entropy bound. The arrays have bounded diameter. A product cover for $S_Z$ and $\mathcal{H}$, the bound $N(S_Z, \|\cdot\|_2, u) \le C(1 + 1/u)^{d_Z}$, and the entropy integral for $\mathcal{H}$ show that their

Dudley integral is at most $C\{1+\sqrt{\log(2\ell_n)}\}$. Conditional symmetrization, Dudley's inequality (Dudley, 1967), and Markov's inequality thus give

$$\sup_{z,h}\left|\alpha_n(z)'\hat{Q}_n^{-1}n^{-1/2}\sum_i \Psi_n(Z_i)e_i[h]\right| = O_p(1+\sqrt{\log(2\ell_n)}).$$

Restricting to the well-conditioned event before taking conditional expectations avoids any moment requirement on $\hat{Q}_n^{-1}$.

For the second process, use projection orthogonality $E[\Psi_n(Z)v_h(Z)] = 0$ and bound its norm by the supremum over $\|u\|_2 \le 1$ of $|n^{-1/2}\sum_i u'\Psi_n(Z_i)v_h(Z_i)|$, times $\|\hat{Q}_n^{-1}\|_{\text{op}}$. After symmetrization, its empirical metric is bounded by

$$CR_n(s)\|\hat{Q}_n^{1/2}\|_{\text{op}}\{M_H\|u-u'\|_2 + \|h-k\|_\infty\}.$$

Use (C.1) and $N(\{u : \|u\|_2 \le 1\}, \|\cdot\|_2, t) \le (1+2/t)^{L_n}$. The resulting entropy integral is bounded by $CR_n(s)\|\hat{Q}_n^{1/2}\|_{\text{op}}(1+\sqrt{L_n})$. The matrix expectation bound also implies $E\|\hat{Q}_n^{1/2}\|_{\text{op}} = O(1)$, so symmetrization and Markov give $O_p(\sqrt{L_n}R_n(s))$ for this process. Multiplying by $\bar{\xi}_n/\sqrt{n}$ and adding the projection bias $C_A M_H R_n(s)$ proves the lemma. ∎

**Proof of Theorem 3.4.** For $h = \Phi_n'\beta \in \mathcal{H}_n$, $\hat{\Pi}_n'\beta = \Psi_n'\hat{\gamma}[h]$ and $\Pi_n'\beta = \mathbb{A}[h]$. The preceding lemma bounds the error uniformly over the larger, fixed class $\mathcal{H}$. Since $\sqrt{L_n}R_n(s) = O(1)$ and $\log\ell_n = O(\log L_n)$, its bound is $O_p(a_n)$ and contains no $K_n$.

For $\hat{g}_n$, apply Belloni et al. (2015), Theorem 4.3. Their conditional error-moment requirement is supplied by the uniform bound on $E[|Y-g_0(Z)|^m \mid Z]$. Use their approximation bound $\ell_k c_k = O(R_n(s))$. After division by $\sqrt{\log(2L_n)}$, the two remainder bounds have orders

$$\sqrt{\frac{L_n\log(2L_n)}{n^{1-2/m}}} + \frac{L_nR_n(s)}{\sqrt{n}}, \qquad R_n(s),$$

respectively. These are bounded under the stated assumptions. Their design, basis, and sample conditions are also satisfied. Consequently $\|\hat{g}_n - g_0\|_\infty = O_p(a_n)$. Adding the two errors proves the claim. Since $L_n\log(2L_n)/n \to 0$ and $L_n^{-v/d_Z} \to 0$, this rate vanishes. ∎

**Proof of Theorem 3.5.** Put $I = [x_-, x_+]$ and $L = x_+ - x_- > 0$. Taylor's formula, evaluated from any point toward the more distant endpoint, gives the common

derivative bound $\|h'\|_\infty \le 2/L + cL =: B_1$ for $h \in \mathcal{H}(c)$. Extend $h$ outside $I$ by its endpoint tangent lines, obtaining $H \in C^{1,1}(\mathbb{R})$ with $\|H'\|_\infty \le B_1$ and $\|H''\|_\infty \le c$ in the weak-derivative sense. Unlike a constant extension, this extension has no jumps in its first derivative at the endpoints.

For $\rho > 0$, define $H_\rho(x) = (2\rho)^{-1} \int_{-\rho}^{\rho} H(x+u)\,du$. Then $\|H_\rho - h\|_{\infty,I} \le B_1\rho$, and

$$H_\rho''(x) = \frac{H'(x+\rho) - H'(x-\rho)}{2\rho}$$

has supremum norm at most $c$ and Lipschitz constant at most $c/\rho$. Use the nonnegative, partition-of-unity B-spline basis $\{B_{j,q-2}\}$ of order $q-2$ on the same mesh to form

$$w_n(x) = \sum_j H_\rho''(\zeta_j) B_{j,q-2}(x),$$

where $\zeta_j$ is any point in the support of the corresponding basis function. Each such support has length at most $C_q\Delta_n$. Therefore

$$\|w_n\|_\infty \le c, \qquad \|w_n - H_\rho''\|_\infty \le C_q c\Delta_n/\rho.$$

These elementary basis properties are described in de Boor (2014). Integrate $w_n$ twice, choosing the integration constants to agree with $H_\rho$ and $H_\rho'$ at $x_-$. The resulting function $p_n$ is an order-$q$ spline in the original sieve, with $\|p_n''\|_\infty \le c$, and

$$\|p_n - h\|_\infty \le B_1\rho + (L^2/2)C_q c\Delta_n/\rho.$$

Take $\rho = \sqrt{\Delta_n}$ and denote the deterministic right-hand bound by $e_n = O(\sqrt{\Delta_n})$. Since $-e_n \le p_n \le 1 + e_n$, the spline

$$q_n[h] = \frac{p_n + e_n}{1 + 2e_n}$$

lies in $[0,1]$, has curvature bounded by $c$, and obeys $\|q_n[h] - h\|_\infty \le 2e_n$. For $q \ge 4$ these splines are $C^2$, so they belong to $\mathcal{H}(c)$. The constant spline $1/2$ has common shape slack at least $\min\{1/2, c\}$ for the restrictions $(h, -h, h'', -h'')' \le (1, 0, c, c)'$.

The common value and derivative bounds make $\mathcal{H}(c)$ uniformly bounded and equicontinuous. Arzelà–Ascoli gives compact closure in $C(I)$. To check the entropy integral directly, choose a mesh of size proportional to $u$ and round the values of $h$ at

its nodes to a grid of size proportional to $u$. Linear interpolation gives a supremum-norm cover with error at most $u$, after adjusting the fixed constants. There are $O(u^{-1})$ nodes and $O(u^{-1})$ possible rounded values at each node, so $\log N(\mathcal{H}(c), \|\cdot\|_\infty, u) \leq Cu^{-1}\log(C/u)$ for $0 < u < 1$. The square root of this bound is integrable at zero. ∎

## C.1 Proofs for Section 5

**Proof of Proposition 5.1.** Taking an infimum over the larger class $\mathcal{H}(c_2)$ cannot increase the value of the objective in (5.1). ∎

**Proof of Theorem 5.2.** Suppress $c$. If $h \in \mathcal{H}$ has population residual at most $b^\star + \epsilon$, its feasible sieve approximation has empirical residual at most $b^\star + \epsilon + \kappa_n + \eta_n$. It is feasible on both grids at this value of the discrepancy variable. Letting $\epsilon \downarrow 0$ gives

$$\hat{b}_n^\star \leq b^\star + \kappa_n + \eta_n.$$

Choose the fixed $B$ in the theorem. Eventually the right side is strictly less than $B$, with probability tending to one, because $b^\star \leq \|g_0 - \mathbb{A}[h^s]\|_\infty < B - 1$. The boundedness argument in the proof of Theorem 3.2, applied at $B$ with anchor $q_n^s$, bounds the norms of all grid-feasible candidates with discrepancy at most $B$ by a fixed constant. This uses only fixed positive slack, $\Theta_B$ bounded, and $\eta_n + d_n = o_p(1)$.

Take a grid candidate with discrepancy $t \leq B$ and objective arbitrarily close to $\hat{b}_n^\star$. Its continuum empirical moment violation is $O(d_n)$ and its shape violation is $O(d_n)$, uniformly over such candidates. Mix it with $q_n^s$ with weight $O(d_n)$ to repair just the shape inequalities, whose anchor slack is at least $\sigma$. The repaired function $h^r$ belongs to $\mathcal{H}_n$. Convexity of the empirical residual norm shows

$$\|\hat{g}_n - \widehat{\mathbb{A}}_n[h^r]\|_\infty \leq t + O(d_n), \qquad \widehat{\mathbb{A}}_n[\Phi_n'\beta] = \hat{\Pi}_n'\beta,$$

because the empirical residual of $q_n^s$ is bounded in probability. Apply the first-stage bound to $h^r \in \mathcal{H}_n$ to obtain $b^\star \leq t + O_p(d_n) + \eta_n$. Let the optimization error tend to zero. Together with the upper bound, this proves $|\hat{b}_n^\star - b^\star| = O_p(r_n)$. ∎

**Proof of Corollary 5.3.** Under the null,

$$P(\varphi_n = 1) \leq P(\hat{b}_n^\star - b^\star > \lambda_n) \longrightarrow 0,$$

since $\hat{b}_n^\star - b^\star = O_p(r_n)$ and $r_n = o(\lambda_n)$. Under a fixed alternative let $\Delta = b^\star - b_0 > 0$. Eventually $\lambda_n < \Delta/2$, and $P(\varphi_n = 0) \leq P(b^\star - \hat{b}_n^\star \geq \Delta/2) \to 0$. These are pointwise-in-distribution statements; uniform size control would require uniform versions of the underlying rate assumptions. ■